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Lab: Creating a Model

In this lab we will create a model for the linear regression problem we started tackling in the previous section (predicting house prices from square feet of living space). This lab is relatively straightforward: we will create a single, simple model as shown in video lectures. For the time being, we will not try out different models, because it is difficult to understand their effectiveness until we train and test the model. We will however become comfortable inspecting the model.

Carry off where you left off at the end of the last lab. Alternatively, there is a reference copy of the code attached as a resource in itsLearning (in this task) for this lesson which you may use.

Refer to the TensorFlow.js API docs as you complete each step.

Part 1 - Defining a model

1. Create a new function which will create a sequential model and return it:

```
function createModel () {
    const model = tf.sequential();
    // More code to go here
    return model;
}
```

2. Using `model.add`, add a single `dense` layer to the model with:

- One input dimension (`inputDim`) as we only have 1 feature.
- One unit (`units`).
- A `linear` activation function.
- A bias (`useBias`).

```
model.add(tf.layers.dense({
    inputDim: 1,
    units: 1,
    activation: 'linear',
    useBias: true,
}));
```

3. Create an optimiser that uses the SGD algorithm, using a learning rate of `0.1`:

a. const optimizer = tf.train.sgd(0.1);
4. Compile the model using Mean Squared Error as the loss function:

```
a. model.compile({
b.    optimizer,
c.    loss: 'meanSquaredError'
d. });
```

5. At the end of the `run` method, add a call to the `createModel` function:

```
a. const model = createModel();
```

Part 2 - Inspecting a model

1. Print a textual summary of the model:

```
a. model.summary();
```

2. Using `tfjs-vis`, show a similar summary of the model on the Visor:

```
a. tfvis.show.modelSummary({ name: `Model Summary`, tab: `Model` }, model);
```

3. Using `tfjs-vis`, show a summary of the layer. As the first (and only) layer, it has index 0.

```
a. tfvis.show.layer({ name: `Layer 1 `, tab: `Model Inspection` }, model.getLayer(undefined, 0));
```

Extra challenge

Add an extra layer to the model and inspect as above. Experiment with different numbers of nodes, and see what happens when the bias is turned off.

- TensorFlow.js API docs
- tsjs-vis API docs
- Solutions for Section 5

linear-section-5-start.html •••